

THEDFORD, NE, USA

WMO: 722211

Lat: 41.964N

Lon: 100.569W

Elev: 2914

StdP: 13.21

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 94087

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-4.3	1.0	-10.5	3.5	-2.3	-5.6	4.5	3.0	32.4	23.4	28.8	21.5	6.1	290	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.4	96.9	67.8	93.0	68.2	90.1	67.6	73.4	86.3	71.8	84.8	70.5	83.1	12.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.0	123.2	79.5	68.2	115.5	77.6	66.4	108.5	76.4	39.1	86.3	37.6	84.6	36.3	82.9	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.4	24.9	21.5	DB	-14.8	101.8	2.3	4.2	-16.4	104.8	-17.7	107.3	-19.0	109.6	-20.6	112.6	(4)
(5)			WB	-15.2	76.0	2.2	1.6	-16.8	77.2	-18.1	78.1	-19.4	79.0	-21.0	80.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	27.4	28.2	40.1	48.0	58.1	69.4	75.6	72.6	64.3	50.2	38.9	27.4	(6)	
	DBStd	19.71	11.98	12.55	11.30	9.82	9.14	6.91	6.07	5.87	8.53	9.49	10.59	11.68	(7)	
	HDD50	3003	700	614	338	149	31	0	0	0	4	113	352	702	(8)	
	HDD65	6307	1164	1031	771	512	251	33	4	11	115	466	782	1167	(9)	
	CDD50	3055	0	3	33	90	280	582	793	700	434	119	21	0	(10)	
	CDD65	884	0	0	1	3	36	165	332	246	95	6	0	0	(11)	
	CDH74	10733	0	1	39	121	586	2010	3930	2626	1231	173	16	0	(12)	
	CDH80	4771	0	0	7	29	195	835	1984	1183	494	43	1	0	(13)	
(14) Wind	WSAvg	9.4	9.2	9.2	10.1	11.3	10.5	9.3	8.5	8.2	9.0	9.4	8.9	8.8	(14)	
(15) Precipitation	PrecAvg	22.0	0.4	0.6	1.1	2.6	3.3	3.7	3.0	2.5	1.9	1.7	0.7	0.4	(15)	
	PrecMax	29.3	1.4	1.8	3.0	6.2	6.9	8.2	7.0	5.0	4.9	5.6	2.2	2.5	(16)	
	PrecMin	9.9	0.0	0.0	0.1	0.4	0.4	0.3	0.6	0.4	0.4	0.1	0.0	0.0	(17)	
	PrecStd	4.4	0.3	0.4	0.7	1.4	1.9	1.7	1.5	1.1	1.1	1.3	0.6	0.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.7	68.1	79.4	84.4	90.7	98.5	102.8	99.0	95.4	85.9	75.9	62.6	(19)	
		MCWB	44.3	47.7	53.1	56.4	63.1	67.8	67.1	67.3	65.5	59.7	52.3	45.2	(20)	
	2%	DB	54.7	59.2	72.0	77.1	85.7	92.8	98.4	94.2	90.7	78.8	68.2	54.2	(21)	
		MCWB	41.5	44.1	50.2	54.3	62.6	68.3	68.2	67.8	66.0	56.6	49.0	41.7	(22)	
	5%	DB	48.4	52.3	65.8	72.0	80.9	88.4	94.4	90.5	86.2	73.0	62.0	48.6	(23)	
		MCWB	38.7	40.3	48.0	53.1	60.5	67.2	69.1	68.2	64.5	54.7	46.3	39.0	(24)	
	10%	DB	43.5	45.9	59.5	66.4	75.9	84.7	90.5	87.2	81.3	67.8	55.9	43.4	(25)	
		MCWB	36.0	37.5	45.8	50.8	59.2	66.2	68.9	68.1	62.4	52.7	44.1	35.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.7	49.3	56.9	59.8	68.4	74.1	76.1	75.1	71.8	62.4	54.3	47.2	(27)	
		MCDB	59.7	66.9	70.4	75.7	82.3	87.3	88.8	87.6	84.3	76.7	70.3	60.8	(28)	
	2%	WB	42.1	44.6	52.9	57.2	65.4	72.1	73.9	72.7	69.0	59.6	50.6	42.5	(29)	
		MCDB	54.2	58.0	66.4	72.0	79.9	85.3	87.0	84.4	82.2	72.6	65.0	53.8	(30)	
	5%	WB	39.2	41.0	49.9	54.7	62.9	70.0	72.2	71.1	67.4	56.6	47.5	39.0	(31)	
		MCDB	47.7	51.1	64.1	68.6	76.2	82.7	85.8	82.6	81.1	69.5	62.1	47.9	(32)	
	10%	WB	36.5	37.7	46.3	52.1	60.6	68.1	70.9	69.6	65.5	53.8	44.2	36.0	(33)	
		MCDB	42.8	45.5	60.0	63.7	72.8	80.9	84.4	80.9	77.7	65.6	55.4	42.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	21.9	22.1	24.9	23.9	23.9	24.6	25.4	25.0	26.9	25.6	24.8	21.8	(35)	
		MCDBR	29.1	30.5	36.0	34.1	31.6	29.9	30.7	30.4	32.2	35.0	33.9	27.9	(36)	
		MCWBR	17.6	17.5	18.1	15.7	13.0	10.8	8.8	10.2	11.6	16.4	17.6	17.2	(37)	
	5% WB	MCDBR	26.5	29.6	32.5	29.8	26.7	26.2	25.6	25.2	27.1	30.4	32.6	27.4	(38)	
		MCWBR	16.7	17.6	17.5	15.1	12.7	11.6	10.4	10.3	11.7	15.7	17.0	17.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.259	0.268	0.292	0.326	0.346	0.365	0.384	0.383	0.342	0.307	0.269	0.260	(40)	
	taud		2.534	2.510	2.477	2.391	2.422	2.398	2.388	2.373	2.454	2.520	2.548	2.517	(41)	
	Ebn at Noon		290	303	305	299	294	288	280	278	283	283	283	279	(42)	
	Edh at Noon		22	26	31	36	36	37	37	36	31	26	21	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		698	981	1325	1647	1939	2217	2223	1912	1549	1080	764	594	(44)	
	RadStd		54	56	103	94	141	146	77	89	85	106	49	43	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+2.34	+2.52	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page